GPT-6 OLI 开环测试报告

0811_candy_v21 Episode 0 三个独立观测帧 2026-09-12

测试目标

检查 GPT-6 的高层 tool call，以及 OpenAIResponsesOliVLA 将该目标线性展开成 50×66 action chunk 后的结果。重点分析任务阶段、手腕平移、手腕旋转和手指命令。全部测试均为 observation-only，没有向机器人发布动作。

任务指令

```
Goal: Pick up the purple candy and place it in the right section of the snack tray with right arm.
```

运行条件

- GPT 原始输出：一个 control_oli tool call，不直接输出 chunk。
- Policy wrapper 输出：[1,50,66]，其中位姿由代码线性插值。
- 每帧独立调用，Action call 为 1/50；不伪造执行历史。
- 手腕和头部平移：base 系相对 dx/dy/dz。
- 不进入 Operator/MROS，不发布机器人动作。

离线数据限制

parquet 中 observation.base_pose 的平移恒为 [0,0,0]，不代表实机底盘高度；数据集没有双脚 observation，因此使用双脚占位值。实机推理由 MROS 提供这些状态。

报告不包含 API key；结果来自实际 Responses API request/response。

完整 System Prompt

以下保留实际发送给 GPT 的英文原文

第 1 部分

```
You control an OLI humanoid through relative
wrist and head motions. At every turn you receive
head, left-wrist, and
right-wrist camera images, the current absolute poses,
 and both hand states.
Use exactly one control_oli tool call. Omitted targets
 hold their current
values.

Infer the current task phase from the images, robot
state, and prior turns.
Continue from the observed state; do not restart a
fixed grasping sequence or
infer the phase from finger closure alone. Make one
small, deliberate move.
Each hand target contains its six native finger
channels in this order:
[thumb flexion, thumb lateral splay, index, middle,
ring, little].
Useful reference profiles are open [0, 98, 0, 0, 0,
0], normal grasp
[30, 92, 40, 98, 98, 98], and firm grasp [58, 92, 58,
98, 98, 98]. Values
between profiles provide partial closure. Channel 1
controls lateral thumb
splay rather than finger flexion, so its open value
intentionally stays near
98. Omit an unused hand to hold all six current
channels. Keep the unused wrist
and head still unless moving them is necessary.
Position units are metres in
the robot base frame: +x points forward from the
robot, +y points to the
robot's left, and +z points upward. Wrist and head
dx/dy/dz commands are
relative displacements in that frame. Wrist rotation
commands are relative to a
right-handed hand frame: local +x points from wrist to
 fingers, local +y lies
laterally across the palm, and local +z is the outward
 palm normal. Roll twists
the wrist about +x, pitch moves the fingers up/down
about +y, and yaw moves the
fingers left/right about +z. Positive angles follow
the right-hand rule. Use
wrist rotation when the visible finger or palm
alignment requires it. Include
a short note explaining the visible evidence and
requested move. Commands are
bounded by the runtime; use the next observation to
verify what was executed.
```

第 2 部分

GPT 控制工具契约

每次 request 携带相同 JSON schema；以下为等价可读形式

control_oli

```
control_oli({
  "targets": {
    "left_wrist"?: {
      "dx"?: number, "dy"?: number, "dz"?: number,
      "rotation_delta_rpy_deg"?: [number, number, number]
    },
    "right_wrist"?: { 与 left_wrist 相同 },
    "head"?: { "dx"?: number, "dy"?: number, "dz"?: number },
    "left_hand"?:  [0..100 共 6 项],
    "right_hand"?: [0..100 共 6 项]
  },
  "note": string
})

手指顺序：
[thumb_flex, thumb_splay, index, middle, ring, little]

参考值：
张开      [0, 98, 0, 0, 0, 0]
普通抓握  [30, 92, 40, 98, 98, 98]
牢固抓握  [58, 92, 58, 98, 98, 98]

省略字段表示保持当前值。
```

原始 request JSON 中保留完整机器可读 schema、三张 JPEG 的 data URL 和所有消息。

50×66 Action Chunk 展开方式

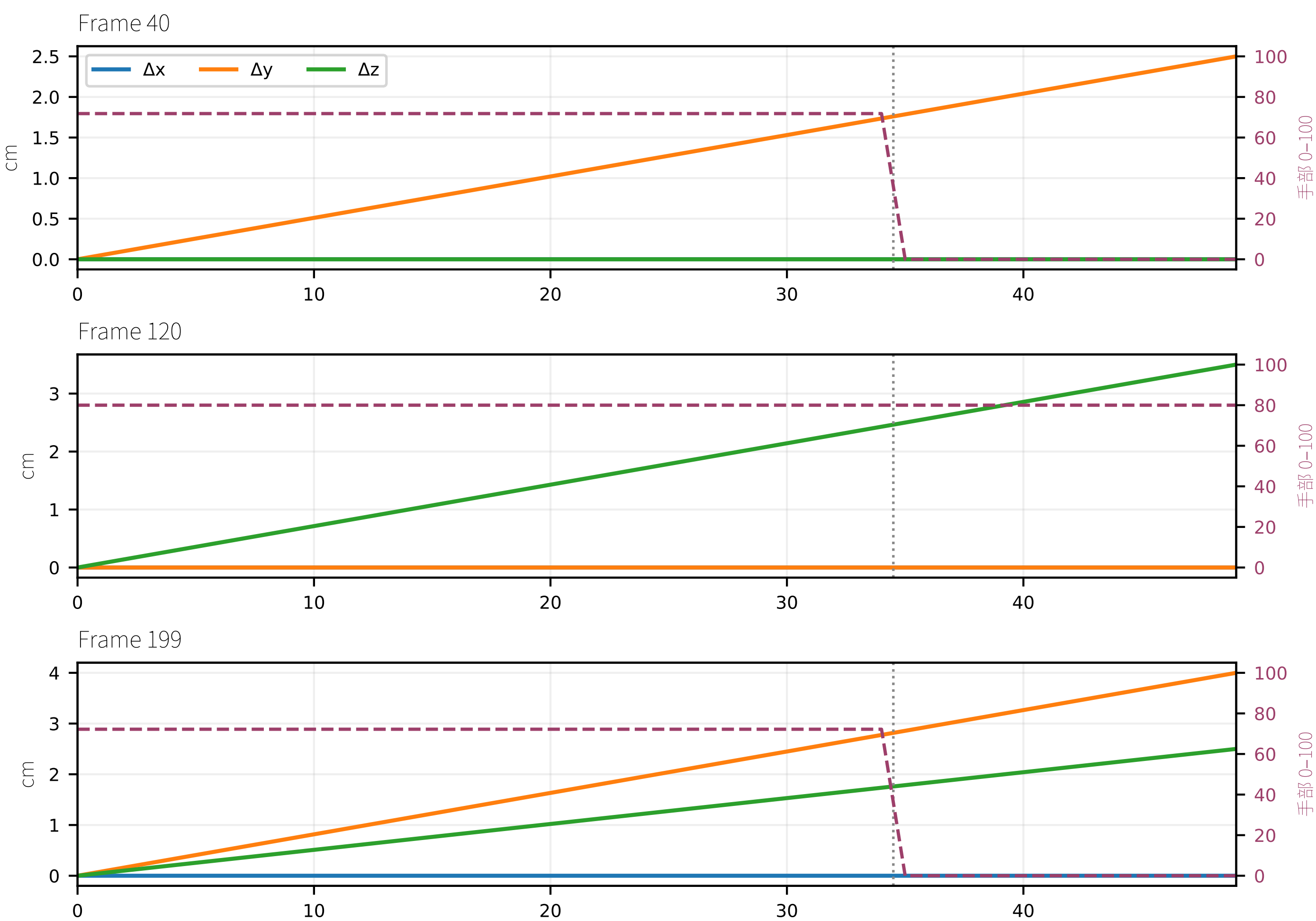
GPT 返回目标；Policy wrapper 在完整 50 点上做线性插值

输出张量

[batch=1, horizon=50, action_dim=66]

时间结构

30 Hz；50 点约 1.67 秒。腕部和头部在第 0–49 点持续做等步长线性插值，最后一点才到达目标；旋转沿相对旋转向量线性推进。手部目标仍在最后 15 点（第 35–49 点）应用。wrapper 不额外生成终点 hold，chunk 结束后的保持由下层控制器负责。



阅读说明

逐帧页面中的“66D 终点”只是第 49 个控制点摘要。GPT 不逐点输出 chunk；中间位姿点由 wrapper 线性生成，终点之后不在本层追加 hold。

Frame 40

独立开环调用　延迟 8.66s　输入/输出 token 2434/157

头部相机

左腕相机

右腕相机

完整文字输入

```
Current OLI observation.
Instruction: Pick up the purple candy and place it in the right section of the
snack tray with right arm.
Action call: 1/50.
Max wrist displacement per call: 0.080 m.
Max head displacement per call: 0.030 m.
Max relative rotation per call: 15.0 degrees.
Workspace xyz bounds: [[-0.3 ,  0.8 ],
 [-0.65,  0.65],
 [-0.3 ,  1.5 ]]
left_wrist xyz in base frame: [0.16357, 0.21532, 0.12475]
right_wrist xyz in base frame: [ 0.22993, -0.17302,  0.05636]
head xyz in base frame: [0.06312, 0.00871, 0.58252]
left_foot xyz in base frame: [0., 0., 0.]
right_foot xyz in base frame: [0., 0., 0.]
base xyz in world frame: [0., 0., 0.]
left_hand [thumb_flex, thumb_splay, index, middle, ring, little]: [ 2., 97.,  2.,
  2.,  2.,  2.]
right_hand [thumb_flex, thumb_splay, index, middle, ring, little]: [29., 92., 39.,
 97., 97., 97.]
```

GPT 完整 tool call


```
{
  "targets": {
    "right_wrist": {
      "dy": 0.025
    },
    "right_hand": [
      0,
      98,
      0,
      0,
      0,
      0
    ]
  },
  "note": "The purple candy is on the tabletop just left of the right hand, not yet grasped. Open the right hand and shift slightly left to begin aligning over it; hold the left arm and head still."
}
```


动作终点与 GT

66D 终点
右腕 xyz: [0.22992999851703644,
-0.14801999926567078,
0.056359998881816864]
头部 xyz: [0.06312000006437302,
0.008709999732673168, 0.5825200080871582]
手部线序 [L0,R0,...]:
[2.0, 0.0, 97.0, 98.0, 2.0, 0.0, 2.0, 0.0,
2.0, 0.0, 2.0, 0.0]

同帧 GT
右腕 delta: [0.02143, 0.01279, -0.00739]
右腕局部 RPY(度): [2.54, -1.26, 6.38]
手部线序 [L0,R0,...]:
[0.0, 30.0, 98.0, 92.0, 0.0, 40.0, 0.0,
98.0, 0.0, 98.0, 0.0, 98.0]

分析

方向上，+y 与 GT 一致；但遗漏了 +x、下降和手腕旋转。GPT 判断为尚未抓取并要求张手，而 GT 已开始收拢手指。

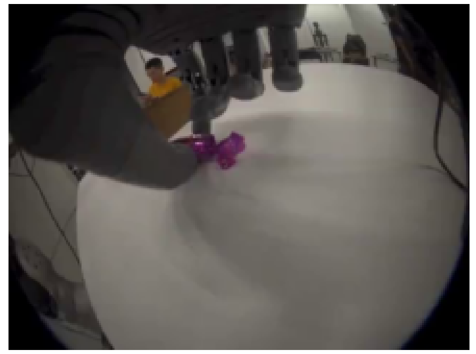
Frame 120

独立开环调用　延迟 9.58s　输入/输出 token 2434/108

头部相机

左腕相机

右腕相机



完整文字输入

```
Current OLI observation.
Instruction: Pick up the purple candy and place it in the right section of the
snack tray with right arm.
Action call: 1/50.
Max wrist displacement per call: 0.080 m.
Max head displacement per call: 0.030 m.
Max relative rotation per call: 15.0 degrees.
Workspace xyz bounds: [[-0.3 ,  0.8 ],
 [-0.65,  0.65],
 [-0.3 ,  1.5 ]]
left_wrist xyz in base frame: [0.16952, 0.2149 , 0.12313]
right_wrist xyz in base frame: [ 0.25183, -0.17172,  0.01647]
head xyz in base frame: [0.06703, 0.00812, 0.58193]
left_foot xyz in base frame: [0., 0., 0.]
right_foot xyz in base frame: [0., 0., 0.]
base xyz in world frame: [0., 0., 0.]
left_hand [thumb_flex, thumb_splay, index, middle, ring, little]: [ 2., 97.,  2.,
  2.,  2.,  2.]
right_hand [thumb_flex, thumb_splay, index, middle, ring, little]: [55., 92., 54.,
 97., 97., 97.]
```

GPT 完整 tool call


```
{
  "targets": {
    "right_wrist": {
      "dz": 0.035
    }
  },
  "note": "The purple candy is beneath the curled right fingers beside the tray. Lift the right wrist slightly while holding the current grip to check that the candy is secured and clear of the table."
}
```


动作终点与 GT

66D 终点
右腕 xyz: [0.25183001160621643,
-0.1717199981212616, 0.05147000029683113]
头部 xyz: [0.06702999770641327,
0.00812000036239624, 0.5819299817085266]
手部线序 [L0,R0,...]:
[2.0, 55.0, 97.0, 92.0, 2.0, 54.0, 2.0,
97.0, 2.0, 97.0, 2.0, 97.0]

同帧 GT
右腕 delta: [0.00444, 0.0403, -0.01042]
右腕局部 RPY(度): [5.26, -0.98, -6.33]
手部线序 [L0,R0,...]:
[0.0, 58.0, 98.0, 92.0, 0.0, 58.0, 0.0,
98.0, 0.0, 98.0, 0.0, 98.0]

分析

保持抓握是正确的，但 GPT 选择上抬。GT 主要沿 +y 搬运，并有轻微下降和约 8.3° 的局部旋转；单帧仍难以稳定区分抓取与搬运阶段。

Frame 199

独立开环调用　延迟 11.92s　输入/输出 token 2434/205

头部相机

左腕相机

右腕相机

完整文字输入

```
Current OLI observation.
Instruction: Pick up the purple candy and place it in the right section of the
snack tray with right arm.
Action call: 1/50.
Max wrist displacement per call: 0.080 m.
Max head displacement per call: 0.030 m.
Max relative rotation per call: 15.0 degrees.
Workspace xyz bounds: [[-0.3 ,  0.8 ],
 [-0.65,  0.65],
 [-0.3 ,  1.5 ]]
left_wrist xyz in base frame: [0.15833, 0.21148, 0.12336]
right_wrist xyz in base frame: [ 0.3022 , -0.10138,  0.0649 ]
head xyz in base frame: [0.06671, 0.00872, 0.58197]
left_foot xyz in base frame: [0., 0., 0.]
right_foot xyz in base frame: [0., 0., 0.]
base xyz in world frame: [0., 0., 0.]
left_hand [thumb_flex, thumb_splay, index, middle, ring, little]: [ 2., 97.,  2.,
  2.,  2.,  2.]
right_hand [thumb_flex, thumb_splay, index, middle, ring, little]: [30., 92., 40.,
 97., 97., 97.]
```

GPT 完整 tool call


```
{
  "targets": {
    "right_wrist": {
      "dy": 0.04,
      "dz": 0.025
    },
    "right_hand": [
      0,
      98,
      0,
      0,
      0,
      0
    ]
  },
  "note": "The purple candy is resting in the tray, left of the empty right hand. Open the hand and shift slightly left and upward to begin aligning above it with clearance."
}
```


动作终点与 GT

66D 终点
右腕 xyz: [0.30219998955726624,
-0.06137999892234802, 0.08990000188350677]
头部 xyz: [0.06671000272035599,
0.00872000027447939, 0.5819699764251709]
手部线序 [L0,R0,...]:
[2.0, 0.0, 97.0, 98.0, 2.0, 0.0, 2.0, 0.0,
 2.0, 0.0, 2.0, 0.0]

同帧 GT
右腕 delta: [-0.00715, 0.02118, 0.01294]
右腕局部 RPY(度): [-0.85, 1.35, -14.65]
手部线序 [L0,R0,...]:
[0.0, 0.0, 98.0, 98.0, 0.0, 0.0, 0.0, 0.0,
 0.0, 0.0, 0.0, 0.0]

分析

三帧中最接近 GT。GPT 与 GT 都给出 +y、+z 和张开右手；主要遗漏轻微 -x 以及约 -14.65° 的局部 yaw。

结论与后续验证

1. 删除固定的“张手→接近→下降→闭合→抬升”流程后，阶段偏置有所降低，但抓取与搬运交界处仍存在单帧歧义。

2. Frame 199 的方向和手指状态最接近 GT；Frame 120 仍将横向搬运判断成上抬。

3. 三次调用均未主动输出手腕旋转，而 GT 使用约 7–15° 的旋转。这是目前最稳定的能力缺口。

4. 位置转换已统一为 target_xyz = current_xyz + delta_xyz，因为 keypoint observation 和命令位移都位于机器人 base 坐标系。

5. 当前数据无法验证底盘平移：其 base translation 恒为零。后续需使用包含实测 base 位姿的数据，或采集 live MROS observation-only trace。

原始 trace 文件

```
frame_40_request.json
frame_40_response.json
frame_40_summary.json
frame_120_request.json
frame_120_response.json
frame_120_summary.json
frame_199_request.json
frame_199_response.json
frame_199_summary.json
```

API 返回的 reasoning.summary 和 reasoning.content 均为空，只有加密字段。因此报告包含全部可见输入和 tool 输出，但不声称能够读取隐藏思维链。